Classical Mechanics: Foundations of Motion

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Chapter 1: Kinematics

Displacement is defined as the change in position of an object in a given direction. Velocity is the rate of change of displacement with respect to time, and is a vector quantity. Speed refers to the magnitude of velocity without direction.

The formula for average velocity is $v = d / t$, where d is displacement and t is time. Acceleration is defined as the rate of change of velocity with respect to time, given by $a = (v2 - v1) / t$.

For example, a car that travels 100 metres in 5 seconds has an average velocity of 20 metres per second.

Chapter 2: Newton's Laws of Motion

Newton's first law states that an object remains at rest or in uniform motion unless acted upon by a net external force. The principle of inertia describes this tendency to resist changes in motion.

Force is defined as the product of mass and acceleration, expressed by the formula $F = m a$. This principle leads to a change in the momentum of the object.

The process of solving a dynamics problem follows clear steps. First, draw a free body diagram. Second, resolve the forces. Third, apply Newton's second law. Finally, solve for the unknown quantity.

For example, a force of 10 newtons applied to a 2 kilogram mass produces an acceleration of 5 metres per second squared. In practice, engineers apply this framework to design safe braking systems for vehicles.

Chapter 3: Work and Energy

Work is defined as the product of force and the displacement in the direction of the force, given by $W = F d$. Energy refers to the capacity of a system to do work.

Kinetic energy is the energy possessed by an object due to its motion, expressed by the formula $KE = 0.5 m v^2$. The principle of conservation of energy states that energy cannot be created or destroyed, only transformed.

For example, a falling object converts potential energy into kinetic energy as it descends.

References

1. Halliday, D. Fundamentals of Physics. 2. Serway, R. Physics for Scientists.
Bibliography and further reading available at the publisher website. Acknowledgements: the author would like to thank the review committee.